

Per-slice paired differences on the full validation set (n = 7,334 slices)

■ $\Delta > 0$: slice favors the replacement
■ $\Delta < 0$: slice favors the baseline

(a) Controlled substitution: SS2D vs. GRU

SSIM

750 – 78.2% favor SS2D

slices

Δ SSIM ($\times 10^{-3}$)

PSNR

1000 – 73.8% favor SS2D

Δ PSNR (dB)

nMSE

2000 – 73.8% favor SS2D

Δ nMSE (%)

L1

1500 – 76.1% favor SS2D

Δ L1

(b) Enhanced SS2D vs. controlled SS2D

SSIM

55.8% favor enhanced

slices

Δ SSIM ($\times 10^{-3}$)

PSNR

54.2% favor enhanced

Δ PSNR (dB)

nMSE

3000 – 54.2% favor enhanced

Δ nMSE (%)

L1

2000 – 54.4% favor enhanced

Δ L1

Signs are oriented so that positive always favors the replacement (nMSE/L1 differences are negated).
x-axes span the 99.5th percentile of $|\Delta|$ per panel; the $\leq 0.5\%$ of slices beyond this range are not shown.